

Validity of Inference Methods for Causal Trees: The Role of Small, Imbalanced Leaves

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1 Introduction

When it comes to detecting subgroups that differ with respect to their treatment effects researchers have to be careful when doing this manually. To avoid spurious manual splits algorithms like tree-based methods that discover treatment effect heterogeneity are a popular alternative. Causal trees have first been introduced by Athey and Imbens [2016] and are based on CART algorithms which are used to detect heterogeneity in outcomes.

The resulting Causal Tree contains several and within each leaf the treatment effect is estimated via an appropriate estimator. An important question for researchers is whether the confidence intervals that are applied on the estimator of treatment effects are asymptotically valid. A necessary condition for this is that the size of the leaves grows over the entire support with the total sample size.

Here, a recent paper from Cattaneo et al. [2025] claims the opposite. Centrally, the authors state that uniform convergence of causal tree estimators is arbitrarily slow. This means that at least for some regions of the covariate space the treatment effect estimator does not converge to its true value as the sample size goes to infinity. The reason for this is seen in splits close to the boundary of the parent node that lead to small, imbalanced child nodes. The problem of such imbalanced splits has been described before (Breiman et al. [1984]) and is also referred to as end-cut preference (Ishwaran [2015]). For the resulting small child nodes the effective sample size as the authors claim may not even increase with the total sample size and a normal distributional approximation not hold. Consequently standard confidence intervals i.e. inference would not be valid.

When introducing causal trees Athey and Imbens [2016] crucially introduced the concept of honesty. By honesty the authors understand that not the same information is used for construction of the tree as for the estimation of treatment effects. This means that the resulting leaves of the tree i.e. subgroups of the data which have the same treatment effects can be treated as exogenously given. The exogeneity was supposed to promote the applicability of standard inference methods within these leaves. However, as Cattaneo et al. [2025] claim, convergence rates for honest estimators do not improve significantly such that the same obstacles for valid inference as before prevail.

The main contribution of this paper shall be to examine the properties of confidence intervals of causal tree estimators in the light of the theoretical results of Cattaneo et al. [2025]. We will compare the performance of adaptive (non-honest) and honest estimators. Furthermore, we will shed light on the behavior of the splitting process and characteristics of leaves in areas of the support where coverage decays.

2 Theoretical Overview

2.1 Estimator of treatment effects

Potential outcomes are defined according to the causal model of Rubin [1974].

$$Y_i^{\text{obs}} = Y_i(W_i) = \begin{cases} Y_i(0) & \text{if } W_i = 0, \\ Y_i(1) & \text{if } W_i = 1. \end{cases} \quad (1)$$

The Conditional Average Treatment Effect (CATE) is defined as the difference of the outcome of treated individuals minus the outcome of untreated individuals conditional on the covariates.

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]. \quad (2)$$

Based on the above definition of CATE we can define our estimator as a difference-in-means (DIM) estimator according to Cattaneo et al. [2025]. For this purpose we use the sample analog of the above definition.

Suppose $\mathbf{T}$ is the tree used, and $\mathcal{D}_\tau = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \dots, n_\tau\}$, with $n_\tau \leq n$, is the dataset used. Let $\mathbf{t}$ be the unique terminal node in $\mathbf{T}$ containing $\mathbf{x} \in \mathcal{X}$.

The Difference-in-Means (DIM) estimator is

$$\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau) = \frac{1}{n_1(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} d_i y_i - \frac{1}{n_0(\mathbf{t})} \sum_{i: \mathbf{x}_i \in \mathbf{t}} (1 - d_i) y_i, \quad (3)$$

where $n_d(\mathbf{t}) = \sum_{i=1}^{n_\tau} \mathbb{1}(\mathbf{x}_i \in \mathbf{t}, d_i = d)$, for $d = 0, 1$, are the “local” sample sizes. We set $\hat{\tau}_{\text{DIM}}(\mathbf{x}; \mathbf{T}, \mathcal{D}_\tau) = 0$ whenever $n_0(\mathbf{t}) = 0$ or $n_1(\mathbf{t}) = 0$.

Since we abstract from selection bias an inverse Probability Weighting (IPW) estimator is not relevant.

2.2 Tree framework

At a broader level a tree is being constructed as follows. The original node of the tree consists of the entire sample and is split into two child nodes which contain a subset of the data. By recursive binary splitting this procedure is continued until a certain stopping criterion is reached. This is either a certain tree depth or a minimum leaf size. The sample has then been split into a multitude of subsets i.e. final nodes to which we also refer as leaves. To separate the data into two child nodes at each parent node of the tree a criterion function is applied ([Hastie et al., 2009, p. 309 ff.]

All observations within a leaf are assigned a Treatment Effect that corresponds to the Average Treatment Effect (ATE) within that leaf. Conditioning on the covariates within that leaf we can refer to this ATE as the Conditional Average Treatment Effect (CATE).

2.3 Criterion function

At each node of the tree the algorithm tries to maximize an objective/criterion function. The goal is to discover as much treatment heterogeneity as possible.

The main part of Athey and Imbens [2016] deals with deriving criteria based on the Mean Squared Error (MSE). We will first focus on the adaptive case i.e. the case where the same sample is used for creating the tree and estimating treatment effects. As usual evaluation of the MSE is done based on test set. The MSE for a given treatment effect $\hat{\tau}$ within a leaf looks as follows.

$$\begin{aligned} \text{MSE}_\tau(\mathcal{S}^{\text{te}}, \mathcal{S}^{\text{tr}}, \Pi) &\equiv \frac{1}{\#(\mathcal{S}^{\text{te}})} \sum_{i \in \mathcal{S}^{\text{te}}} \left\{ (\tau_i - \hat{\tau}(X_i; \mathcal{S}^{\text{tr}}, \Pi))^2 \right\} \\ &= \frac{1}{\#(\mathcal{S}^{\text{te}})} \sum_{i \in \mathcal{S}^{\text{te}}} (\tau_i^2 - 2\tau_i \hat{\tau}(X_i; \mathcal{S}^{\text{tr}}, \Pi) + \hat{\tau}^2(X_i; \mathcal{S}^{\text{tr}}, \Pi)) \end{aligned} \quad (4)$$

Problematically, this criterion is infeasible because we do not observe the counterfactual outcome for individuals and hence we also do not observe the individual treatment effects τ_i . This is also referred to as the fundamental problem of causal inference (Holland [1986]). Therefore, some adjustments of the above criterion are necessary.

When formulating their MSE criterion the authors adjust the Expected Mean Squared Error (EMSE) by $-E[\tau_i^2]$ which does not depend on X_i and hence does not affect how the splits are made. The authors also divide by $N^{tr} = \#(\mathcal{S}^{tr})$. Once the number of elements in the training and test split is set dividing by N^{tr} or N^{te} does not change which split Π minimizes the error.

However, we still need an estimator for τ_i in $-2\tau_i\hat{\tau}(X_i; \mathcal{S}^{tr}, \Pi)$. For that we can use $\hat{\tau}_i$ as an unbiased estimator such that $\mathbb{E}_{\mathcal{S}^{te}}[\tau_i \mid i \in \mathcal{S}^{te} : i \in \ell(x, \Pi)] = \mathbb{E}_{\mathcal{S}^{te}}[\hat{\tau}(x; \mathcal{S}^{te}, \Pi)]$. By applying the sample analogs we receive the following adjusted MSE:

$$\begin{aligned} & \widehat{\text{MSE}}_{\tau}(\mathcal{S}^{te}, \mathcal{S}^{tr}, \Pi) \\ &= -\frac{2}{N^{tr}} \sum_{i \in \mathcal{S}^{te}} \hat{\tau}(X_i; \mathcal{S}^{te}, \Pi) \cdot \hat{\tau}(X_i; \mathcal{S}^{tr}, \Pi) + \frac{1}{N^{tr}} \sum_{i \in \mathcal{S}^{te}} \hat{\tau}^2(X_i; \mathcal{S}^{tr}, \Pi) \end{aligned}$$

Instead of evaluating the MSE using a test set we can evaluate the error in-sample which leads to $-\widehat{\text{MSE}}_{\tau}(\mathcal{S}^{tr}, \mathcal{S}^{tr}, \Pi)$ as the criterion.

$$-\widehat{\text{MSE}}_{\tau}(\mathcal{S}^{tr}, \mathcal{S}^{tr}, \Pi) = \frac{1}{N^{tr}} \sum_{i \in \mathcal{S}^{tr}} \hat{\tau}^2(X_i; \mathcal{S}^{tr}, \Pi) \quad (5)$$

From this criterion function we can directly derive what Cattaneo et al. [2025] call variance maximization. When $-\widehat{\text{MSE}}_{\tau}(\mathcal{S}^{tr}, \mathcal{S}^{tr}, \Pi)$ is maximized and tries to find the best split we essentially maximize the variance of the estimator between the child nodes. This is done by having as different as possible treatment effects in the nodes since we sum up squared treatment effects. The algorithm maximizes the criterion on each node and not the whole tree at once, it is greedy. Hence, when to decide where to make a split we want to get the sum of the criterion over both child nodes as large as possible. Said differently we want to explain as much heterogeneity in the treatment effects as possible by splitting one group with one treatment effect into two different groups with two treatment effects. The two child nodes L, R have a summed up error of

$$\text{SST}_{\text{after}} = \sum_{i \in t_L} \hat{\tau}_L^2 + \sum_{i \in t_R} \hat{\tau}_R^2 = n(t_L)\hat{\tau}_L^2 + n(t_R)\hat{\tau}_R^2$$

However, where and if to make the split should be evaluated with respect to the variance/heterogeneity in τ_i that is already explained in the parent node. We can express the error in the parent node in terms of the two errors in the child node.

$$\text{SS}_{\text{before}} = \sum_{i \in t} \hat{\tau}_t^2 = n(t)\hat{\tau}_t^2 = n(t)\left(\frac{n(t_L)\hat{\tau}_L + n(t_R)\hat{\tau}_R}{n(t)}\right)^2$$

Subtracting $\hat{\tau}_t$ from $\text{SST}_{\text{after}}$ yields

$$\begin{aligned}
\text{Gain} &= [n(t_L)\hat{\tau}_L^2 + n(t_R)\hat{\tau}_R^2] - n(t)\hat{\tau}_t^2 \\
&= n(t_L)\hat{\tau}_L^2 + n(t_R)\hat{\tau}_R^2 - n(t) \left(\frac{n(t_L)\hat{\tau}_L + n(t_R)\hat{\tau}_R}{n(t)} \right)^2 \\
&= n(t_L)\hat{\tau}_L^2 + n(t_R)\hat{\tau}_R^2 - \frac{(n(t_L)\hat{\tau}_L + n(t_R)\hat{\tau}_R)^2}{n(t)} \\
&= \frac{n(t_L)n(t_R)}{n(t)} (\hat{\tau}_L - \hat{\tau}_R)^2
\end{aligned} \tag{6}$$

The last step is achieved through algebraic simplification by expanding the square and creating a common denominator. This yields that the criterion which is supposed to be maximized can be written as a weighted difference of the squared treatment effects.

2.4 Honesty

In the paper from Athey and Imbens [2016] the authors centrally apply a procedure which they call honesty. It means to perform the splitting of features on one part of the sample and to estimate treatment effects on the other part of the sample. The downside of this is reduced efficiency since the tree is built on a lower sample size. The advantage is that estimation of treatment effects is now exogenous. This implies that the algorithm does not risk to fit the tree based on spurious extreme values (noise). Hence, the authors argue that the approach eliminates bias.

The splitting criterion is modified and considers that the estimates are now unbiased. Moreover, the authors derive a penalty term that is supposed to prevent high variances in leaf estimates that stem from small leaf sizes from only using part of the sample for estimation.

To derive the modified criterion we first rewrite the original MSE criterion and take the expectation to receive the Expected Mean Squared Error (EMSE).

$$\begin{aligned}
-\text{EMSE}_\tau(\Pi) &= -\mathbb{E}_{(\tau_i, X_i), \mathcal{S}^{\text{est}}} [(\tau_i - \tau(X_i; \Pi))^2 - \tau_i^2] \\
&\quad - \mathbb{E}_{X_i, \mathcal{S}^{\text{est}}} [(\hat{\tau}(X_i; \mathcal{S}^{\text{est}}, \Pi) - \tau(X_i; \Pi))^2] \\
&= \mathbb{E}_{X_i} [\tau^2(X_i; \Pi)] - \mathbb{E}_{\mathcal{S}^{\text{est}}, X_i} [\mathbb{V}(\hat{\tau}^2(X_i; \mathcal{S}^{\text{est}}, \Pi))],
\end{aligned} \tag{7}$$

As an estimator for the within-leaf variance $\mathbb{V}(\hat{\tau}^2(X_i; \mathcal{S}^{\text{est}}, \Pi))$ from the estimation sample we can estimate its equivalent on the training sample. This ensures that we do not violate the honesty assumption by using data from the estimation sample. However, since we want to minimize the variance of the final leaves which will be a function of the estimation sample we still scale with the number of samples in the leaf of the estimation sample.

$$\hat{\mathbb{V}}(\hat{\tau}(x; \mathcal{S}^{\text{est}}, \Pi)) \equiv \frac{S_{\mathcal{S}^{\text{tr}}}^2(\ell(x; \Pi))}{N^{\text{est}}(\ell(x; \Pi))}$$

To compute the variance over all leaves we have to weight with the proportion each leaf has compared to the total sample size of the estimation sample which is $\frac{N^{\text{est}}(\ell)}{N^{\text{est}}}$.

$$\begin{aligned}
\hat{\mathbb{E}} \left[\hat{\mathbb{V}}(\hat{\tau}(x; \mathcal{S}^{\text{est}}, \Pi)) \right] &= \sum_{l \in \Pi} \left(\frac{N^{\text{est}}(l)}{N^{\text{est}}} \right) \cdot \left(\frac{S_{\mathcal{S}^{\text{tr}}}^2(l)}{N^{\text{est}}(l)} \right) \\
&= \sum_{l \in \Pi} \frac{S_{\mathcal{S}^{\text{tr}}}^2(l)}{N^{\text{est}}} \\
&= \frac{1}{N^{\text{est}}} \sum_{l \in \Pi} S_{\mathcal{S}^{\text{tr}}}^2(l)
\end{aligned} \tag{8}$$

Furthermore, we need an estimator for $\mathbb{E}_{X_i} [\tau^2(X_i; \Pi)]$. From the definition of the variance we know that $\text{Var}(\hat{\tau}) = \mathbb{E}[\hat{\tau}^2] - (\mathbb{E}[\hat{\tau}])^2$ and since $\hat{\tau}$ is an unbiased estimator of τ^2 we have $(\mathbb{E}[\hat{\tau}])^2 = \tau^2$. It follows

$$\tau^2 = \mathbb{E}[\hat{\tau}^2] - \text{Var}(\hat{\tau})$$

Because τ^2 is the true population mean and thus a fixed constant we have $\tau^2 = E[\tau^2]$. As an estimator for $\mathbb{E}[\hat{\tau}^2]$ we use the estimated squared treatment effect from the training sample and as an estimator of $\text{Var}(\hat{\tau})$ we use the estimated variance from the training sample. This yields

$$\hat{\mathbb{E}} [\tau^2(x; \Pi)] = \hat{\tau}^2(x; \mathcal{S}^{\text{tr}}, \Pi) - \frac{S_{\mathcal{S}^{\text{tr}}}^2(\ell(x; \Pi))}{N^{\text{tr}}(\ell(x; \Pi))}. \tag{9}$$

Combining the estimators for $\hat{\mathbb{V}}(\hat{\tau}(x; \mathcal{S}^{\text{est}}, \Pi))$ and $\mathbb{E}_{X_i} [\tau^2(X_i; \Pi)]$ yields as the modified criterion for honest splitting

$$\begin{aligned}
-\widehat{\text{EMSE}}_{\tau}(\mathcal{S}^{\text{tr}}, N^{\text{est}}, \Pi) &\equiv \frac{1}{N^{\text{tr}}} \sum_{i \in \mathcal{S}^{\text{tr}}} \hat{\tau}^2(X_i; \mathcal{S}^{\text{tr}}, \Pi) \\
&\quad - \left(\frac{1}{N^{\text{tr}}} + \frac{1}{N^{\text{est}}} \right) \cdot \sum_{\ell \in \Pi} S_{\mathcal{S}^{\text{tr}}}^2(\ell(x; \Pi)).
\end{aligned} \tag{10}$$

We see that differing to the criterion for adaptive estimation we have

$$\left(\frac{1}{N^{\text{tr}}} + \frac{1}{N^{\text{est}}} \right) \cdot \sum_{\ell \in \Pi} S_{\mathcal{S}^{\text{tr}}}^2(\ell(x; \Pi))$$

as a penalty term to prevent overfitting. The term prevents the tree to grow too deeply since then the variance term would increase and the criterion not be maximized.

2.5 Cross-Validation

An essential part of Athey and Imbens [2016] is to apply cross-validation to prevent overfitting. Such cross-validation would lead to splitting D_{struct} further to estimate a pruning parameter. However, since we are concerned about imbalanced splits which are supposed to occur with or without cross-validation we abstract from it following the approach in Cattaneo et al. [2025].

3 Main Contributions of Cattaneo et al. [2025]

3.1 Assumptions on the data generating process

The authors formulate several conditions on the data generating process under which their results are supposed to hold.

Assumption 1 (Data Generating Process). $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : 1 \leq i \leq n\}$ is a random sample, where $y_i = d_i y_i(1) + (1 - d_i) y_i(0)$, $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,p})^\top$, and the following conditions hold for all $d = 0, 1$ and $i = 1, 2, \dots, n$.

- (i) $(y_i(0), y_i(1), \mathbf{x}_i) \perp d_i$, and $\xi = \mathbb{P}(d_i = 1) \in (0, 1)$.
- (ii) $y_i(d) = \mu_d(\mathbf{x}_i) + \varepsilon_i(d)$, with $\mathbb{E}[\varepsilon_i(d) | \mathbf{x}_i] = 0$ and $\mathbf{x}_i \perp \varepsilon_i(d)$.
- (iii) $\mu_d(\mathbf{x}) = c_d$ for all $\mathbf{x} \in \mathcal{X}$, where c_d is some constant and $\mathcal{X}$ is the support of $\mathbf{x}_i$.
- (iv) $x_{i,1}, \dots, x_{i,p}$ are independent and continuously distributed.
- (v) There exists $\alpha > 0$ such that $\mathbb{E}[\exp(\lambda \varepsilon_i(d))] < \infty$ for all $|\lambda| < 1/\alpha$ and $\mathbb{E}[\varepsilon_i^2(d)] > 0$.

Assumption (i) expresses the independence of treatment with covariates and potential outcomes and hence assumes a randomized experiment. Besides that we enforce an overlap assumption i.e. a positive probability for every individual of getting the treatment. In (ii) the authors assume a standard regression model with additive, exogenous and independent errors. Assumption (iii) assumes a constant treatment effect. This assumption is certainly the most crucial one. While it may not hold in reality it allows us to keep the model simple. The simplicity is characterized by the fact that under such a constant treatment effect we would expect the splits to occur at the same frequency over the whole support. Any deviations from that indicate a sub-optimal splitting process. Moreover, such a constant treatment effect model is as the authors claim a canonical member of any interesting class of data generating processes because the constant function belongs to all classical smoothness function classes, as well as to the set of functions with bounded total variation. From this it follows that the model under Assumption 1 allows us to define lower bounds on the worst-case probability of a more than marginal deviation of the estimator of $\tau(x)$ from its actual value. Let $\mathbb{P}_1$ be the probability measure under Assumption 1 and $\mathbb{P}$ be a probability measure in any data generating class $\mathcal{D}$ that includes $\mathbb{P}_1$. Then, we have the following inequality.

$$\sup_{\mathbb{P} \in \mathcal{D}} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| > \epsilon \right) \geq \mathbb{P}_1 \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| > \epsilon \right) \quad (12)$$

The constant treatment effect assumption also removes issues related to standard (boundary or other smoothing) bias in nonparametric. Assumption (iv) postulates that the features x are independent and continuously distributed. Importantly, trees are invariant with respect to monotone transformations (Wager and Walther [2015], Cattaneo et al. [2025]). Every random variable can be monotonically transformed by its own cumulative distribution function (cdf) and by the universality of the uniform every cdf follows a uniform. This implies that we can just assume that the covariates are uniformly distributed. Assumption (v) further states that potential outcome errors are sub-exponential, or equivalently, they satisfy a Bernstein moment condition.

3.2 Uniform convergence

The authors define as their central contribution a theorem on the convergence properties of causal tree estimators. While parametric estimators converge with a rate of $\sqrt{n}$, non-parametric estimators like in the given case typically converge at a much slower rate (Hansen [2009]).

The central theorem of the authors regarding uniform convergence looks as follows:

Theorem 1 (Uniform Accuracy). *Suppose Assumption 1 holds, and the underlying causal tree has at least one split (i.e., at least two terminal nodes). Then, for $l \in \{\text{DIM}, \text{IPW}, \text{SSE}\}$ and all $b \in (0, 1)$,*

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_l^{\text{NSS}}(\mathbf{x}) - \tau(\mathbf{x})| \geq C_1 n^{-b/2} \sqrt{\log \log n} \right) \geq b/e,$$

where the positive constant C_1 only depends on the distribution of $(\varepsilon_i(0), \varepsilon_i(1), d_i)$, and (13)

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} |\hat{\tau}_l^{\text{HON}}(\mathbf{x}) - \tau(\mathbf{x})| \geq C_2 n^{-b/2} \right) \geq C_3 b,$$

where the positive constants C_2 and C_3 only depend on the distribution of $(\varepsilon_i(0), \varepsilon_i(1), d_i)$, and the sample splitting scheme via $\liminf_{n \rightarrow \infty} \frac{n_{\tau}}{n}$ and $\limsup_{n \rightarrow \infty} \frac{n_{\tau}}{n}$. The precise definitions of the constants are given in the supplemental appendix of Cattaneo et al. [2025].

The first equation defines lower bounds on the uniform convergence for adaptive estimators and the second equation defines uniform convergence for honest estimators. Since C_1 , C_2 and C_3 are positive constants and hence do not depend on n they do not affect the rate of convergence of the estimators. For the adaptive case we have a rate of convergence which is not faster than $n^{-b/2} \sqrt{\log \log n}$ and for the honest case a rate of convergence not faster than $n^{-b/2}$. Since $b \in (0, 1)$ it holds that $n^{-b/2} > \sqrt{n}$ and hence the convergence rate is slower than in a parametric case.

3.3 Probability of extreme splits

The central reason for the slow uniform convergence is shown by Cattaneo et al. [2025] in an equation that characterizes the probability of extreme splits. This equation also explains the meaning of n^{-b} in Theorem 1. The authors provide their argument based on a decision tree of depth 1 (a so called decision stump). They show as follows that with non-vanishing probability the first split will concentrate near a boundary of the parent node.

Let $\hat{i}$ be the index at which the split is performed (CART split index) and $\hat{j}$ be the split variable at the root node. Further, $l \in \{\text{DIM}\}$, noticing that the first split coincides for no-sample-splitting and honest constructions. For each $a, b \in (0, 1)$ with $a < b$ and $j \in \{1, 2, \dots, p\}$ we have

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \mathbb{P} (n^a \leq \hat{i} \leq n^b, \hat{j} = j) \\ &= \liminf_{n \rightarrow \infty} \mathbb{P} (n - n^b \leq \hat{i} \leq n - n^a, \hat{j} = j) \geq \frac{b - a}{2pe} \end{aligned} \tag{14}$$

Since we want to illustrate the probability of small, imbalanced child nodes, we can let $a \rightarrow 0$. In that case, for relatively small b ,

$$\mathbb{P}(n^a \leq \hat{i} \leq n^b, \hat{j} = j) \geq \frac{b}{2pe}$$

is the probability of a split on the left side of covariates j . Similarly,

$$\mathbb{P}(n - n^b \leq \hat{i} \leq n - n^a, \hat{j} = j) \geq \frac{b}{2pe}$$

is the probability of a split on the right side of covariate j . A split can at the same time occur to the left or to the right of the data and thus the two probabilities are disjoint. Adding up the two probabilities results again for relatively small b in the following lower bound probability.

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"}) = \mathbb{P}(\hat{i} \leq n^b \vee n - n^b \leq \hat{i}) \geq \frac{b}{pe} \quad (15)$$

As an illustration we can choose $n = 100.000$, $a = 0.6$. This results to

$$\inf \mathbb{P}(\hat{i} \leq 100.000^{0.6} = 1000 \vee 100.000 - 100.000^{0.6} = 99000 \leq \hat{i}) \gtrsim \frac{0.2208}{p}$$

We see that the probability of small child nodes decays inversely linear to p . Hence, while for one covariate the probability is large it decreases rapidly as the number of covariates increases. This is the case because at each node a split occurs only along values of a single covariate (Wager and Walther [2015]). Consequently, the probability that a specific variable is chosen at an arbitrary split is $\frac{1}{p}$.

The authors also show that the issue of an imbalanced split is inherited from decisions stumps to trees of arbitrary depth [Cattaneo et al., 2025, app. p. 33]). Due to recursive splitting the K -th parent node satisfies for the optimal split index $\hat{i}$ and $a \rightarrow 0$ that

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\hat{i} \leq n^b) = \liminf_{n \rightarrow \infty} \mathbb{P}(n - n^b \leq \hat{i}) \geq \frac{b}{2pe} \quad (16)$$

Accordingly, the K -th parent node satisfies

$$P(\text{"imbalanced split"}) \geq \frac{b}{pe}$$

We can derive the limit probability that at least one imbalanced split occurs in a tree of depth k . The probability of at least one extreme split can be rewritten as follows

$$\begin{aligned} \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"at least 1 imbalanced split"}) &= 1 - \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"no imbalanced split"}) \\ &= 1 - \left(1 - \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"})\right)^k \end{aligned}$$

Since

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"}) \geq \frac{b}{pe} \Leftrightarrow 1 - \liminf_{n \rightarrow \infty} \mathbb{P}(\text{"imbalanced split"}) \leq 1 - \frac{b}{pe}$$

it holds

$$\liminf_{n \rightarrow \infty} \mathbb{P}(\text{"at least 1 imbalanced split"}) \geq 1 - \left(1 - \frac{b}{pe}\right)^k \quad (17)$$

This result is extensive. It means that for every additional tree depth the probability of having at least one imbalanced split increases.

4 Tree Calibration

When it comes to determining the parameters of the model the key metric is the bias-variance tradeoff which is determined by the magnitude of the parameters i.e. the flexibility of the model [James et al., 2023, p. 32 ff.]. A more flexible model typically has a higher variance while a less flexible model has a higher bias. In our case a more flexible model is a deeper tree which has smaller leaves i.e. a higher variance of the CATE estimates but a smaller bias.

4.1 α - regularity

A popular assumption is to force child nodes to contain at least a share α of data points contained in the parent node, so called α - regularity (Wager and Walther [2015], Wager and Athey [2018]). However, such a condition makes splits along the edges of noise variables less desirable (Wager and Walther [2015]). Since we want to examine imbalanced splits and not rule them per sé we will not impose α - regularity.

4.2 Tree depth vs. minimum leaf size

The two key parameters that define tree depth and need to be adjusted in the simulation part are tree depth and minimum sample size per leaf. There is clearly a relationship between these parameters. A deeper tree tends to have smaller leaf sizes and smaller minimum leaf sizes permit deeper trees. However, if trees as elaborated above may get stuck in small leaves after a few splits fixing the minimum leaf size has a different effect than deciding on a particular tree depth. In the first case tree growth would just stop when a certain sample size is reached in only one leaf while in the second case the tree could still keep growing if the parameter permits so.

Because we are interested in detecting as many extremely small leaves as possible we do not want to stop as soon as minimum leaf size in one leaf appears. Hence, we will set the minimum leaf size equal to 1 and will only deal with the parameter tree depth. However, from the fact that we want to construct confidence intervals for each leaf results an important limitation. The calculation of a variance for the treatment and the control group requires that both groups contain at least two observations. Hence, we will only evaluate leaves that contain at least two treatment group observations and two control group observations.

4.3 Determinants of tree depth

There are several factors that determine tree depth. In the context of Machine Learning an excessively high hyperparameter like tree depth is referred to as overfitting which needs to be prevented. The tree size governs the model's complexity and the optimal tree size should be adaptively chosen

from the data (Hastie et al. [2009]) In general more complex models may lead to overfitting of the data ([James et al., 2023, p. 22]). Usually, a grown tree is pruned by applying a cost-complexity criterion via cross-validation.

4.3.1 Number of covariates

In general, the more complex the data, the more complex the model should be. For us this may indicate that we choose a deeper tree.

4.3.2 Total sample size

A higher total sample size permits to choose a deeper tree since the leaves will be larger on average.

4.3.3 Honesty

Honesty means that only half of the data is used to create the structure of the tree. Compared to using the whole data this implies a smaller tree depth.

4.3.4 Treatment effect heterogeneity

Cattaneo et al. [2025] rely in their prove on a constant treatment effect because as they say such a model "is a canonical member of any interesting class of data generating processes. Therefore, in our simulation we will also consider such a model. In an actual real world setting where we would like to get accurate estimates of the then constant treatment effect we would of course decide to renounce on any split and just compute treatment effects based on the whole dataset. In our case, however, where we want to determine in what way small leaves occur such an approach would be counterproductive. If there is no problem with unbalanced splits and slow convergence the causal tree would still detect the true treatment effect. Hence, treatment effect heterogeneity or the absence of it does not affect our parameter choice.

From this we can derive the probability that at least one imbalanced split has occurred after K splits.

5 Simulation

We are interested in the behavior of confidence intervals of leaves that are located in the tails of the distribution. This brings up two problems. First, if we want to examine the coverage of confidence intervals around the bounds of the distribution we can naturally only come back to data on one site of the bound since for the other site of the bound the support is not defined. This reduces the number of available leaves. Second, we would like to examine the behavior of bins that are as close as reasonable to the boundary and thus as narrow as possible. Both facts imply that we need a high number of Monte Carlo simulations to dispose of enough data within the bins. This aligns with the literature. Attempts with 10.000 Monte Carlo simulations have suffered from results in the tails that were driven by sampling variation. Hence, I have decided to move forward with 25.000 simulations.

5.1 $n = 1000$ **and** $p = 1$

Coverage does not decay further after a tree depth of 7 is reached. This implies that the minimum leaf size is 1 is reached for a tree depth of 1 and the tree does not keep growing for higher tree depths.

5.2 $n = 5000$ **and** $p = 1$

5.3 $n = 1000$ **and** $p = 3$

References

- Susan Athey and Guido Imbens. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360, 2016.
- Leo Breiman, Jerome Friedman, Richard A Olshen, and Charles J Stone. *Classification and regression trees*. Chapman and Hall/CRC, 1984.
- Matias D Cattaneo, Jason M Klusowski, and Ruiqi Rae Yu. The honest truth about causal trees: Accuracy limits for heterogeneous treatment effect estimation. *arXiv preprint arXiv:2509.11381*, 2025.
- Bruce E Hansen. Lecture notes on nonparametrics. *Lecture notes*, 2009.
- Trevor Hastie, Robert Tibshirani, Jerome Friedman, et al. The elements of statistical learning, 2009.
- Paul W Holland. Statistics and causal inference. *Journal of the American statistical Association*, 81(396):945–960, 1986.
- Hemant Ishwaran. The effect of splitting on random forests. *Machine learning*, 99(1):75–118, 2015.
- Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. An introduction to statistical learning with applications in r. 2023.
- Donald B Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology*, 66(5):688, 1974.
- Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.
- Stefan Wager and Guenther Walther. Adaptive concentration of regression trees, with application to random forests. *arXiv preprint arXiv:1503.06388*, 2015.

A Coverage Ratios

A.1 $n = 1000$ and $p = 1$

Coverage depending on leaf mean by depth (bin width = 0.5)

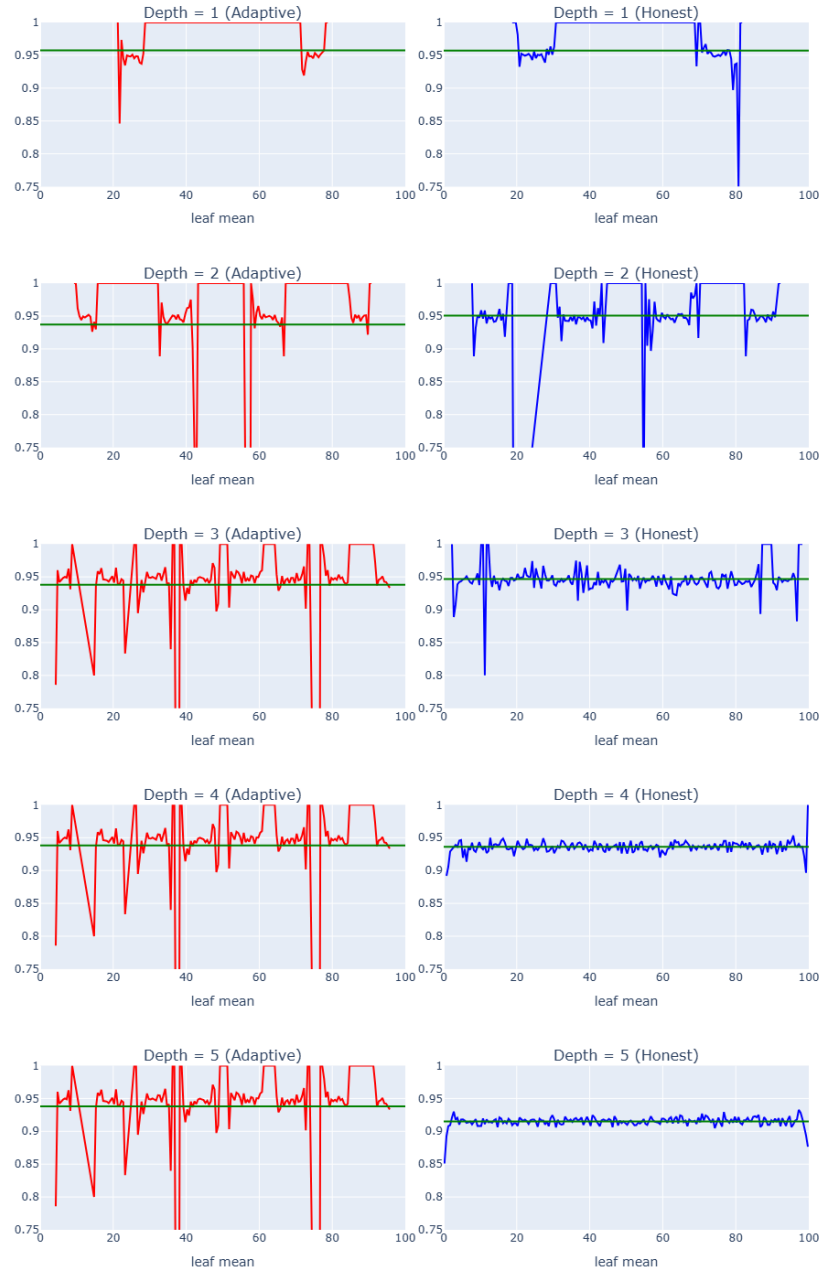


Figure 1: Coverage ratios for the whole support of the covariate ($n=1000$, depth 1-5).

Coverage depending on leaf mean by depth (bin width = 0.5)

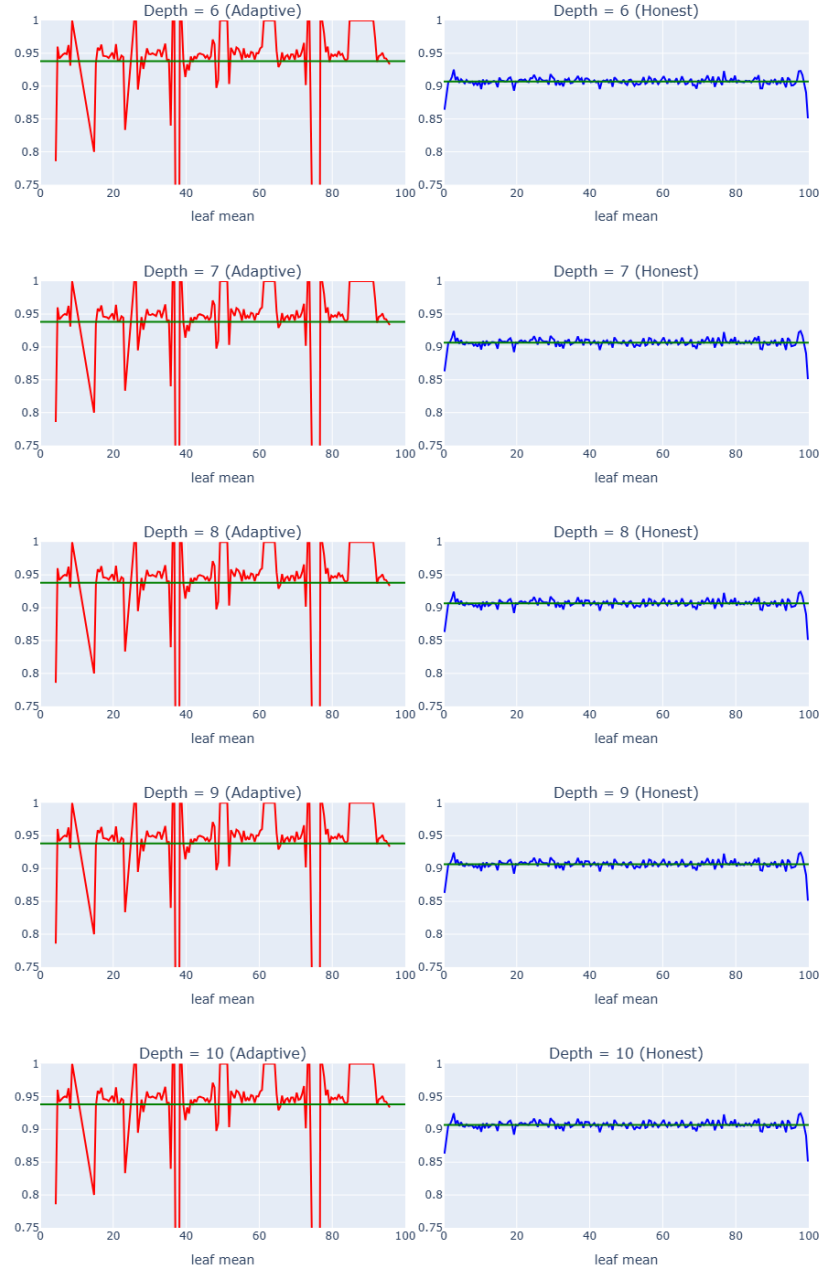


Figure 2: Coverage ratios for the whole support of the covariate ($n=1000$, depth 6-10).

Coverage depending on absolute distance from population mean by depth (bin width = 0.1)

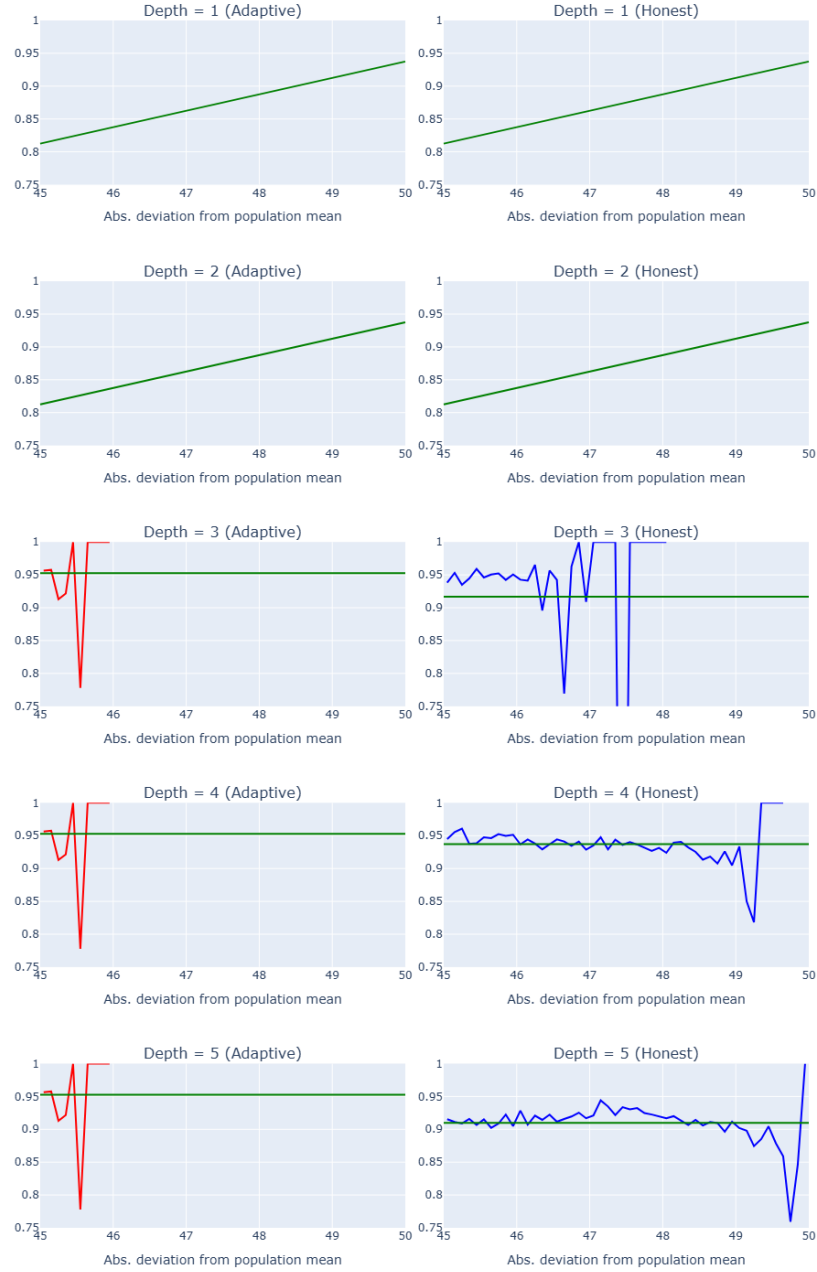


Figure 3: Coverage ratios for absolute deviations from leaf means ($n=1000$, depth 1-5).

Coverage depending on absolute distance from population mean by depth (bin width = 0.1)

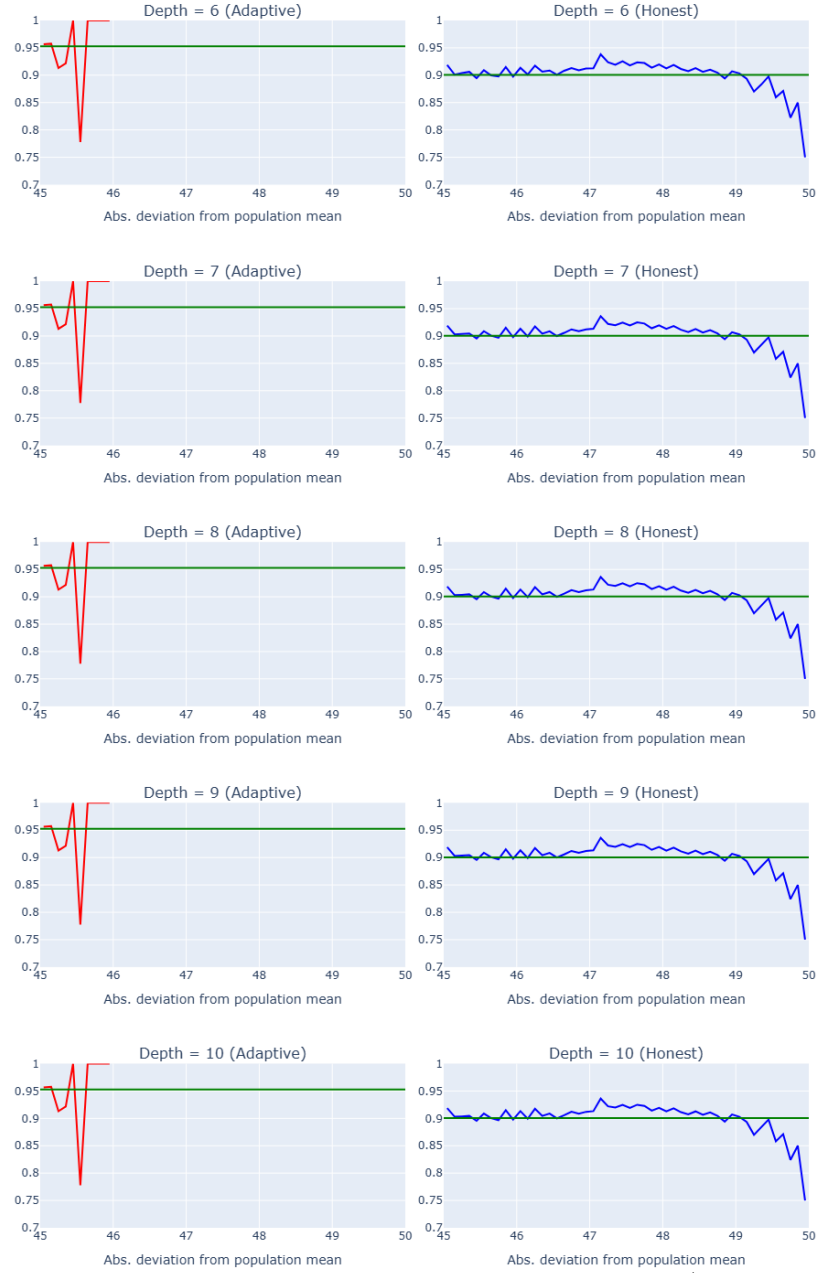


Figure 4: Coverage ratios for absolute deviations from leaf means ($n=1000$, depth 6-10).

Coverage depending on leaf mean by depth (bin width = 0.5)

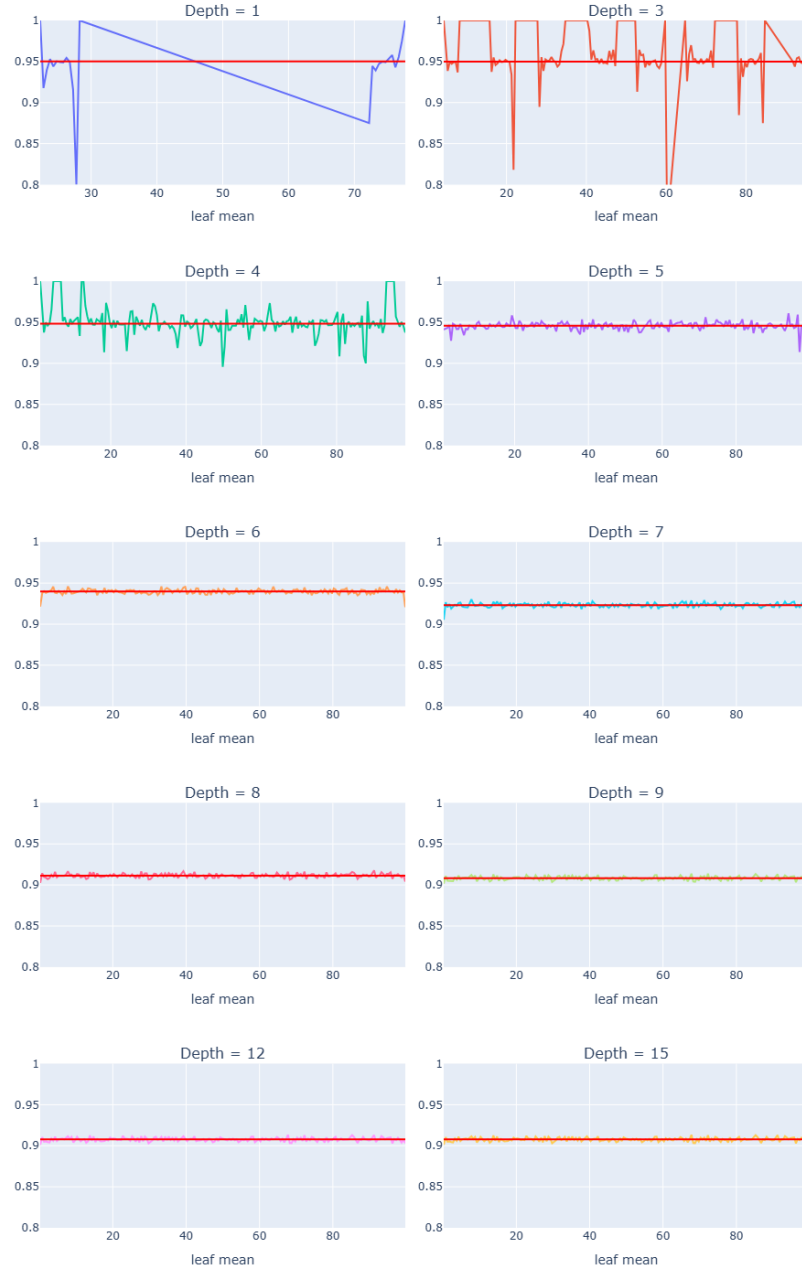


Figure 5: Coverage ratios for the whole support of the covariate ($n=5000$).

Coverage depending on absolute distance from poulation mean by depth (bin width = 0.1)

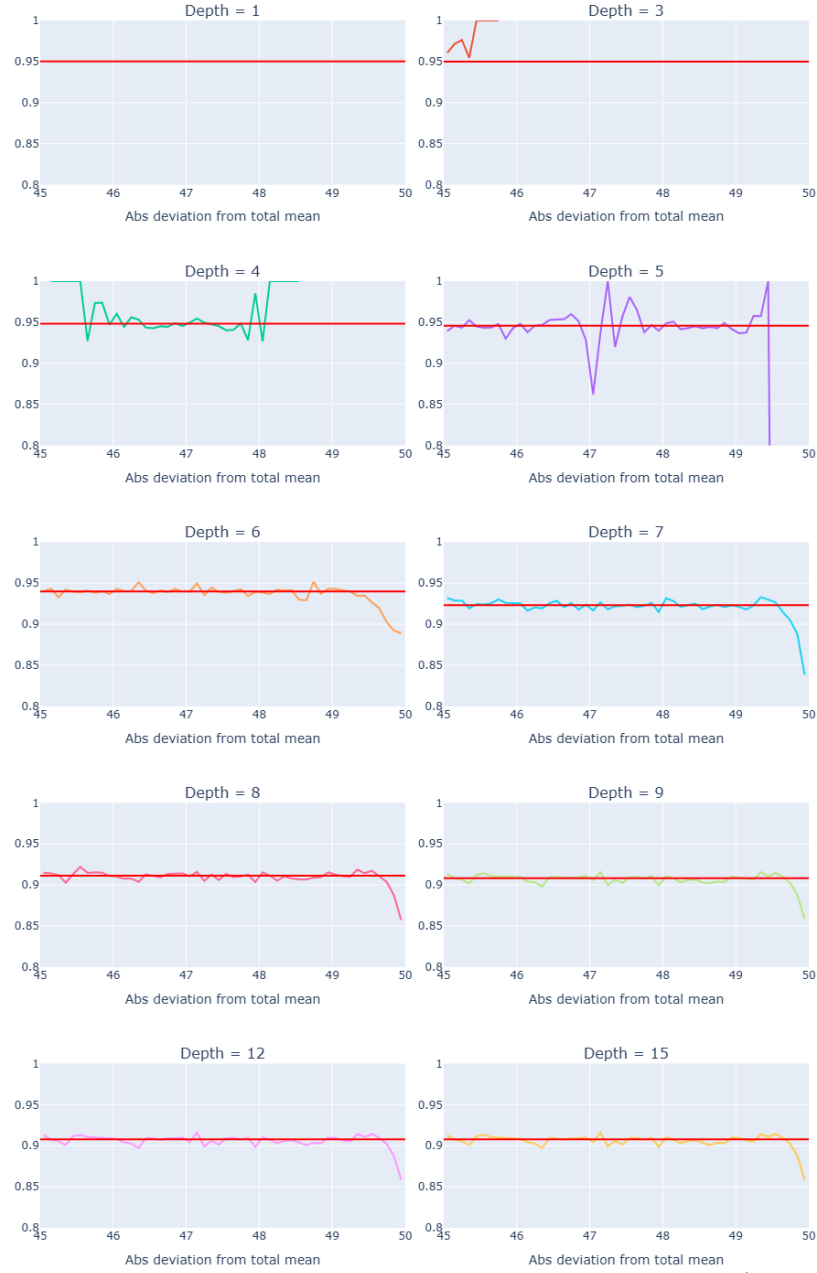


Figure 6: Coverage ratios for absolute deviations of leaf means ($n=5000$).

B Tables for Mean Leaf Sizes

Depth	Mean Leaf Size	
	Coverage	
1	0	249.79
	1	250.01
2	0	124.97
	1	125.00
3	0	62.28
	1	62.51
4	0	30.62
	1	31.30
5	0	15.69
	1	17.21
6	0	13.36
	1	15.08

Table 1: Mean leaf size overall ($n = 1000$)

Depth	Mean Leaf Size	
	Coverage	
1	0	1250.95
	1	1249.95
3	0	311.96
	1	312.53
4	0	156.33
	1	156.25
5	0	77.89
	1	78.14
6	0	38.12
	1	39.14
7	0	18.97
	1	21.07

Table 2: Mean leaf size overall ($n = 5000$)

Abs. Deviation Interval	Depth	Coverage	Mean Leaf Size
(47.499, 48.0]	3	1	33.75
	4	0	23.77
		1	24.52
	5	0	19.78
		1	20.65
	6	0	17.43
(48.0, 48.5]		1	19.02
	4	0	19.41
		1	20.42
	5	0	16.50
		1	17.15
	6	0	15.67
(48.5, 49.0]		1	16.67
	4	0	15.48
		1	16.39
	5	0	12.46
		1	13.38
	6	0	12.19
(49.0, 49.5]		1	13.09
	4	0	10.75
		1	12.03
	5	0	9.29
		1	9.84
	6	0	9.02
(49.5, 50.0]		1	9.58
	4	1	5.00
	5	0	6.42
		1	6.98
	6	0	6.39
		1	6.82

Table 3: Mean leaf size for extreme absolute deviations ($n = 1000$)

Abs. Deviation Interval	Depth	Coverage	Mean Leaf Size
(47.499, 48.0]	4	0	126.14
		1	125.73
	5	0	100.88
		1	101.31
	6	0	38.74
		1	39.89
	7	0	19.78
		1	21.35
(48.0, 48.5]	4	0	107.67
		1	106.89
	5	0	83.19
		1	83.66
	6	0	36.63
		1	37.82
	7	0	19.65
		1	21.79
(48.5, 49.0]	4	1	99.00
		0	66.97
	5	1	67.27
		0	47.12
	6	1	49.19
		0	20.05
	7	1	22.28
		0	22.28
(49.0, 49.5]	5	0	47.66
		1	48.87
	6	0	36.49
		1	36.96
	7	0	20.89
		1	23.04
(49.5, 50.0]	6	0	19.94
		1	21.14
	7	0	14.12
		1	15.62

Table 4: Mean leaf size for extreme absolute deviations ($n = 5000$)

C Leaf and Sample Distributions



Figure 7: Average number of samples per absolute deviations.

Relative Frequency Distribution of Number of Leaves per Leaf Mean (bin width = 0.5)



Figure 8: Share of number of leaves per absolute deviations.

Relative Frequency Distribution of Number of Samples per Leaf Mean (bin width = 0.5)



Figure 9: Share of number of samples per absolute deviations.